Junjie Qiu

MEMORY MODELING / MULTIMODAL / PHYSICAL AI

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01 Personal Profile

I am a first-year Ph.D. student at HKUST (GZ), studying **memory modeling for physical AI**. I aim to develop long-horizon memory for agents in dynamic environments, including embodied systems. My experience combines temporal-model design, pretraining and evaluation with multimodal video generation, cross-shot consistency and application-level narrative memory. I co-authored a NeurIPS 2025 Spotlight paper. Previously at LuckyShort, I was an Architect reporting directly to the CTO. I also worked on temporal-model adaptation at Huawei and HPC systems at SUSTech's Center for Computational Science and Engineering.

02 Education

Hong Kong University of Science and Technology (GZ) Sep. 2026 - Present

Ph.D. in Intelligent Transportation

Southern University of Science and Technology Sep. 2021 - Jul. 2025

B.S. in Data Science and Big Data Technology

GPA: 3.89/4.00 · Rank: 4/53 (top 10%)

Honors: Outstanding Graduate; Outstanding Undergraduate Thesis;
Top Ten Graduate Award, Shude Residential College (Jun. 2025)

03 Industry Experience

ByFusion · LuckyShort Mar. 2026 - Sep. 2026

Architect · AI short-drama production

- Led video-agent architecture for long-form storytelling, coordinating narrative planning, visual reference assets, controllable generation, and multimodal evaluation across an end-to-end production workflow.
- Designed application-level memory for long-form scripts to track character and narrative state, check events against prior context, and update downstream context after revisions for episode continuity.
- Integrated agent-driven script production, video generation, publishing, and ad delivery for a platform reaching 3B+ TikTok/YouTube views, including multiple short-drama videos with 100M+ views each.
- Built preference records linking generated videos to their context, user selections, and rejection reasons, preserving multimodal feedback for visual-quality evaluation and cross-shot temporal consistency checks.

Huawei · 2012 Laboratories Mar. 2025 - Dec. 2025

Research Intern

- Adapted pretrained temporal models to Huawei Ascend, preparing telecom and power time series for model training and evaluation across heterogeneous industrial forecasting datasets.
- Reproduced forecasting baselines and evaluated temporal models on Huawei's internal telecom and power datasets, validating the strongest forecasting results among the methods tested.
- Contributed to FactoST [1]; the paper reports MAE reductions of 12.5% on Electricity vs. OpenCity and 13.7% on Weather vs. Moirai, with 10% training data and 96-step context and forecast horizons.

SUSTech · Center for Computational Science and Engineering Mar. 2024 - Jul. 2024

Intern · High-Performance Computing

- Conducted HPL (LINPACK) and HPCG benchmarking on the Qiming and Taiyi supercomputers, establishing performance baselines for computational throughput and distributed execution efficiency.
- Provisioned and commissioned HPC compute nodes, covering hardware assembly, software deployment, and system debugging to integrate new computing capacity into existing supercomputing clusters.
- Profiled MPI communication to identify bottlenecks, then optimized switch interconnect topology for four nodes installed in Jun. 2024, reducing measured MPI communication time by approximately 20%.

04 Research Experience

FactoST-v2 · Universal Temporal Pretraining Oct. 2025 - Jan. 2026

UTP method design and experiments · Preprint [2]

- Designed and pretrained an encoder-only quantile model on normalized sequence patches with randomly masked history prefixes, configured for up to 2,048 input steps and 256 forecast steps across domains.
- Evaluated fixed-weight zero-shot forecasts on benchmarks excluded from pretraining ($12 \to 12$, $96 \to 96$); masking reduced PEMS08 MAE by 15% vs. no masking (107.22 to 91.09 at $96 \to 96$; Tables IX-X in [2]).

FactoST · Cross-domain Temporal Pretraining Feb. 2025 - May 2025

Cross-domain transfer evaluation · NeurIPS Spotlight [1]

- Evaluated zero-shot and few-shot transfer across diverse forecasting datasets, assessing cross-domain generalization and data-efficient adaptation through the reuse of pretrained temporal representations.

05 Technical Projects

Agentic Video Generation · Cross-Shot Consistency Jul. 2026 - Sep. 2026

ByFusion · LuckyShort · Internal project

- Built reference-conditioned keyframe generation with Gemini 3.8 Flash checks for identity and scene drift, routing feedback to regenerate faulty shots while retaining accepted frames and shared visual references.
- Centralized creative direction across generation stages and refined it with final-video feedback, warm-starting candidate search from model priors to guide exploration of narrative and visual alternatives.

Rare-Earth Pricing · Action-Conditioned Replanning Jan. 2026 - Mar. 2026

China Mobile Research Institute · China Rare Earth Group

- Contributed to long-horizon price modeling and experiments, conditioning forecasts on geopolitical and supply-chain observations and pricing actions to compare market scenarios and support replanning.

Tencent Cloud · Behavior-Aware Resource Allocation May 2025 - Aug. 2025

Research collaboration · Tencent Rhino-Bird Fund

- Contributed to temporal modeling for Tencent Cloud resource allocation, using behavioral profiles and historical allocation sequences as decision context for user-specific compute resource assignment.

ASC24 · LLM Inference Infrastructure & Optimization Apr. 2024

Ranked 10th among 300+ teams · First Prize & Group Competition Award

- Built distributed LLM serving on four nodes with two GPUs per node, integrating vLLM inference engines, Ray orchestration, and RDMA networking for concurrent execution under competition constraints.
- Implemented dynamic request scheduling across Ray-managed vLLM replicas, dispatching tasks to idle instances through an available-worker queue to balance workloads on a fixed GPU cluster.

06 Skills

Physical AI	Concepts: Diffusion / Rectified Flow, Multimodal Conditioning, DreamerV3, JEPA, RSSM, Latent Dynamics, Latent Imagination, Model-based RL, Action Abstraction, Steerable VLA Policies.
Memory & Context	Concepts: Belief-State Modeling, Multimodal Memory, Event-centric Memory, Prospective Memory.
Training, Inference & Simulation	Project work: vLLM, Ray, RDMA, Huawei Ascend. Familiarity: 4D (DP/TP/PP/CP), Continuous Batching, MuJoCo.

07 Bibliography

[1] S. Zhong, J. Qiu, Y. Wu, X. Zou, Z. Rao, B. Yang, C. Guo, H. Xu, and Y. Liang, "Learning to Factorize Spatio-Temporal Foundation Models," in Advances in Neural Information Processing Systems (NeurIPS), 2025. **Spotlight**.

[2] S. Zhong, J. Qiu, Y. Wu, Y. Liu, Y. He, Z. Rao, B. Yang, C. Guo, H. Xu, and Y. Liang, "Learning to Factorize and Adapt: A Versatile Approach Toward Universal Spatio-Temporal Foundation Models." arXiv preprint arXiv:2601.12083, 2026.